ChE 321 HW 3 Lindsey Meisch

Sunday, September 14, 2025 9:14 PM

1.) 4-1) Yes, it would still be correct because water does not appear in the reaction stoichiometry or rate law and its concentration is constant and just provides the solvent medium.

4-2) a) $X = 0.2$ $\quad \Theta_B = \frac{3}{10} = 0.3$

$$C_D = (10)\left(\frac{0.2}{3}\right) = 0.667\ mol/L$$
$$= 0.67\ mol/dm^3$$

$$C_B = 10\left(0.3 - \frac{0.2}{3}\right) = 10(0.3 - 0.667)$$
$$= 2.33\ mol/dm^3$$

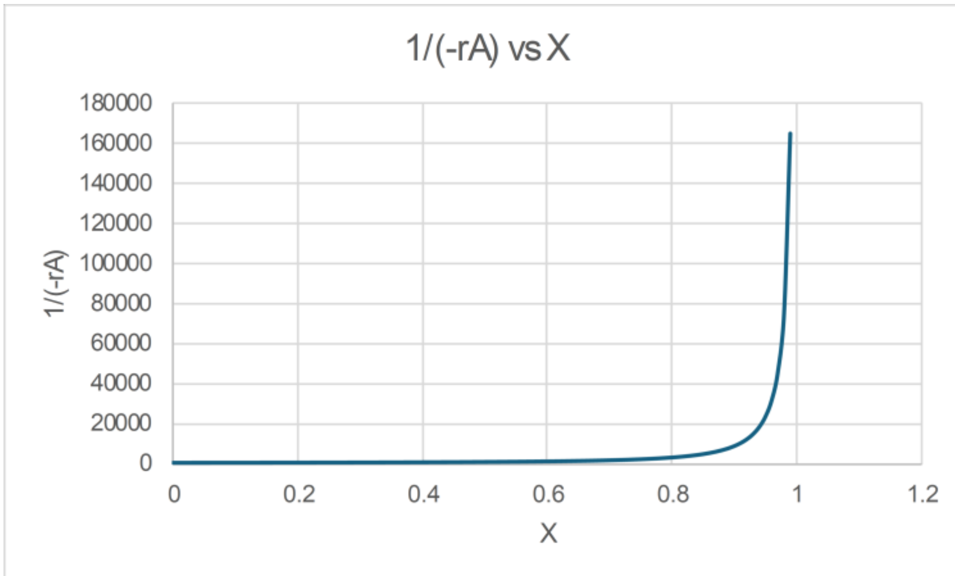
b) $X = 0.9$

$$C_D = 10\left(\frac{0.9}{3}\right) = 3\ mol/dm^3$$

$$C_B = 10\left(0.3 - \frac{0.9}{3}\right) = 10(0.3 - 0.3)$$
$$= 0\ mol/dm^3$$

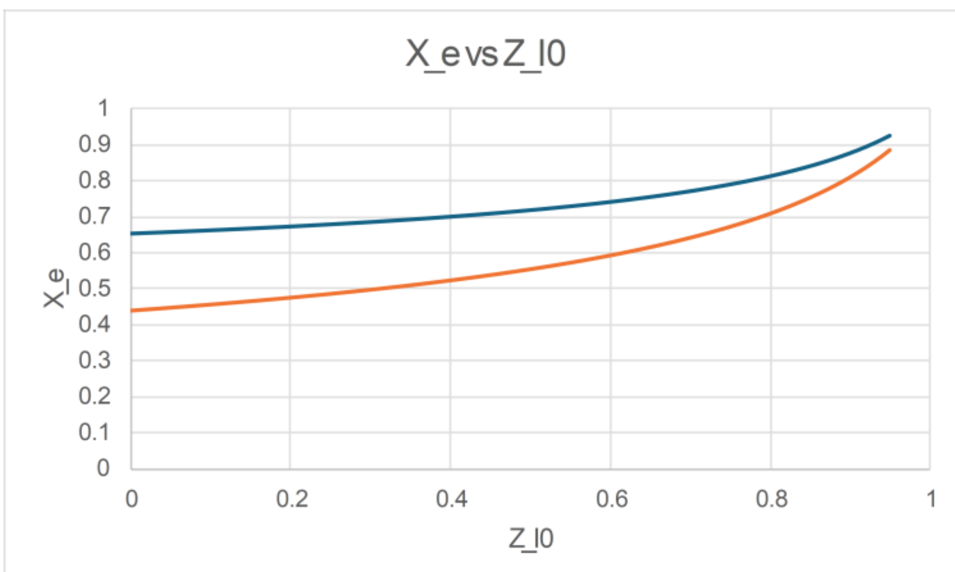
4-3) Reaction basis: SO2 + 1/2 O2 -> SO3

When eps=0, the inert concentration will be constant. So there has to be no net change in the total stoichiometric moles. Since eps=(y_A0)(delta v), delta v would need to be zero.



Observation: the curve grows slowly at a small X, then accelerates and sharply increases near X=1. The rate seems to slow down as X increases and the sharp vertical at the end shows we can never quite reach 100% conversion in a flow reactor because the volume requirement goes to infinity.

4-4) The reaction is creating moles. The flow for PFR/CSTR is fixed at constant T and P, so equilibrium depends on mole fractions. For constant-volume batch, the total moles increase as the reaction moves forward, which raises the total concentration inside the vessel. When the change in stoichiometric moles is greater than 0, equilibrium conversion is higher for the flow system than the batch system.



Where the blue line = flow curve and orange line = batch curve

(1) With delta v > 0 the flow value is above batch. The magnitude of the gap grows with the magnitude of eps, which, for fixed stoichiometry is larger when the feed is concentrated. Larger C_A0 increases the difference. A moderate K_c gives the largest gap.
(2) As the mixture becomes very dilute, the equations converge and the two X values become nearly identical. Large inert fractions make the batch and flow equilibria the closest.

(4-5)

a) $K_A = 0.5 \, min^{-1}$

$C_{A0} = 0.072 \, mol \, dm^{-3}$

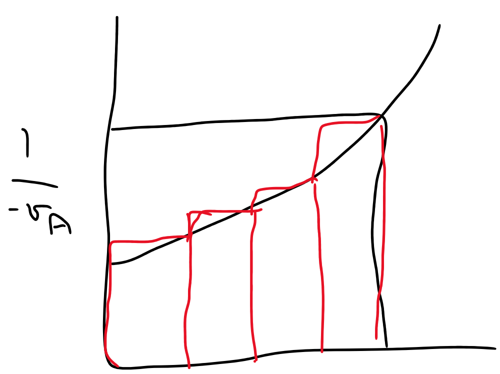
$K_C = 0.1 \, mol \, dm^{-3}$

PFR

$$V = \int_0^X \frac{F_{A0}}{-r_A(X)} \, dX$$

$F_{A0} = 3 \, mol \, min^{-1}$

$X = 0.4$



$$A = 0.1 \left[\frac{39 + 150}{2} + 41 + 60 + 90 \right.$$

$$= 0.1 (94.5 + 191)$$

$$= 28.5$$

$$V = 3(28.5)$$

$$= 85.5\,dm^3$$

$$= 0.0857\,m^3$$

b) $$W = \int_0^X \frac{F_{A0}}{-r'_A(X)}\,dX$$

$$-r'_A(X) = k\,\frac{P_{SO_2}\sqrt{P_{SO_2}} - P_{SO_3}/K_p}{\left(1 + K_{O_2}\sqrt{P_{O_2}} + K_{SO_2}P_{SO_2}\right)^2}$$

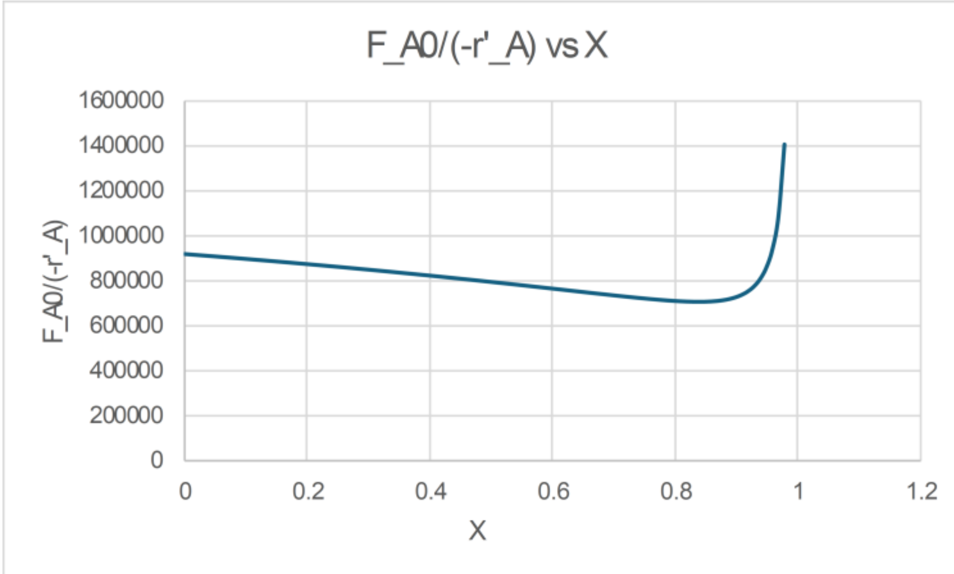
$k = 9.7$

$K_{O_2} = 38.5$ $\quad K_{SO_2} = 42.5$ $\quad K_p = 930$

$P_0 = 14.85$ $\quad y_{A0} = 0.28 \Rightarrow P_{SO_2,0} = 4.158$

$\Theta = 0.54$ $\quad \varepsilon = -0.14$

$F_{A0} = 1000$



For X=0.30, W = 265 kg
For X=0.40, W = 348 kg
For X= 0.99Xe, W=807 kg

2.) a) $n_{A0} = n_{B0}$ $\quad X$ of A

$$n_A = n_{A0}(1-X)$$
$$n_B = n_{B0} - n_{A0}X = n_{A0}(1-X)$$
$$n_C = 8n_{A0}X$$

$$N_T = n_A + n_B + n_C = 2n_{A0}(1-X) + 8n_{A0}X$$
$$= 2n_{A0}(1+3X)$$

$$PV = nRT$$

$$V = 0.1P \Rightarrow 0.1P^2 = N_T RT$$

$$V = \sqrt{0.1 N_T RT}$$

@ $X = 0$

@ $X = 0$

$$v_0 = \sqrt{0.1 N_{T0} RT}$$

$$\therefore \ v(X) = v_0\sqrt{1+3X}$$

$$C_A = \frac{n_A}{v} = \frac{n_{A0}}{v_0}\frac{1-X}{\sqrt{1+3X}}$$

$$= C_{A0}\frac{1-X}{\sqrt{1+3X}}$$

$$C_B = C_{A0}\frac{1-X}{\sqrt{1+3X}}$$

$$-r_A(X) = k_1 C_A^2 C_B = k_1 C_{A0}^3 \frac{(1-X)^3}{(1+3X)^{3/2}}$$

$$N_{T0} = \frac{P_0 V_0}{RT} = \frac{(1.5)(0.15)}{(0.73)(600)} = 5.14\times10^{-4}$$
lbmol

$$n_{A0} = N_{T0}/2$$
$$= 2.57\times 10^{-4}$$

$$C_{A0} = \frac{n_{A0}}{v_0} = \frac{2.57\times10^{-4}}{0.15}$$

$$C_{A0} = \frac{F_{A0}}{v_0} = \frac{\quad}{0.15}$$

$$= 1.71 \times 10^{-3}$$

lb mol/ft^3

$$k_1 C_{A0}^3 = 1(1.71 \times 10^{-3})^3$$

$$\approx \boxed{5.03 \times 10^{-9} \ \frac{\text{lbmol}}{\text{ft}^3 \text{s}}}$$

$$-r_A(X) = 5.03 \times 10^{-9} \frac{(1-X)^3}{(1+3X)^{3/2}}$$

b) $v(X) = V_0 \sqrt{1+3X}$

$$0.2 = 0.15\sqrt{1+3X}$$

$$\Rightarrow \sqrt{1+3X} = \frac{0.2}{0.15} = \frac{4}{3}$$

$$\Rightarrow 1+3X = \frac{16}{9}$$

$$\Rightarrow X = 0.259$$

$$-r_A = 5.03 \times 10^{-9} \frac{(1-0.259)^3}{(1+3(0.259))^{3/2}}$$

$$= 5.03 \times 10^{-9} \frac{0.741^3}{(1.777)^{3/2}}$$

$\approx 8.63 \times 10^{-10}$